MA388 Sabermetrics: Lesson 18

Comparing Peak Trajectories - Part I

LTC Jim Pleuss

Review

What is a *fixed effect*?

What is a *random effect*?

In general, when would you use each?

Career Trajectories

What is a career trajectory?

Why are people interested in them?

We'll start by looking at OPS (on base percentage plus slugging). Here are the relevant equations:

On base percentage:

$OBP = \frac{H+BB+HBP}{AB+BB+HBP+SF}$

(Note: Sacrifice flies (SF) were not recorded until the 1950s.)

Slugging percentage:

$SLG = \frac{(H-X2B-X3B-HR)+2\times X2B+3\times X3B+4\times HR}{AB}$

On base percentage plus slugging:

$OPS = OBP + SLG$

Why do baseball analysts like OPS?

A convenient model for OPS as a function of age is a quadratic function:

$OPS = A + B(Age - 30) + C(Age - 30)^2$

Why use a model (instead of just the data)?

What are some good reasons to use this quadratic model?

What are some bad reasons to use the quadratic model?

Why do we subtract 30 from *Age* in the model?

Based on this model, find an expression for the peak age (where OPS is the highest).

Based on this model, find an expression for the peak OPS.

In the quadratic model, explain what A tells us about a player's trajectory.

In the quadratic model, explain what C tells us about a player's trajectory.

Henry Aaron

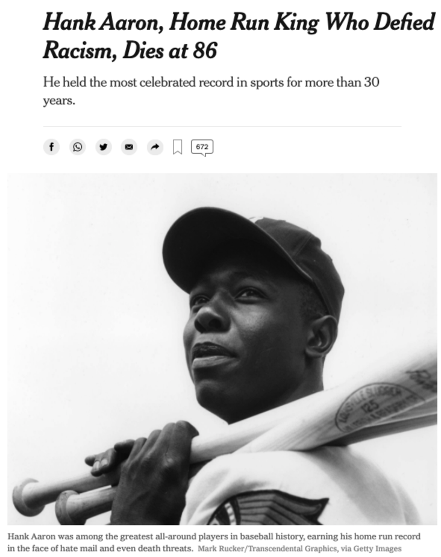


Figure 1: Henry Aaron (Source: *New York Times*)

Let's fit Henry Aaron's trajectory. The `get_stats()` function from the text (Section 8.2) is helpful.

```
library(tidyverse)
library(Lahman)
library(knitr)

# Find his playerID in the People table.
aaronID <- People |>
  filter(nameLast == "Aaron",
         nameFirst == "Hank") |>
  pull(playerID)

# Function to calculate age and OPS.
get_stats <- function(player){
  Batting |>
    filter(playerID == player) |>
    inner_join(People, by = "playerID") |>
    mutate(birthyear = ifelse(birthMonth >= 7,
                              birthYear + 1, birthYear),
           Age = yearID - birthyear,
           SLG = (H - X2B - X3B - HR +
```

```
                    2 * X2B + 3 * X3B + 4 * HR) / AB,
           OBP = (H + BB + HBP) / (AB + BB + HBP + SF),
           OPS = SLG + OBP) |>
    select(playerID, Age, SLG, OBP, OPS)
}
```

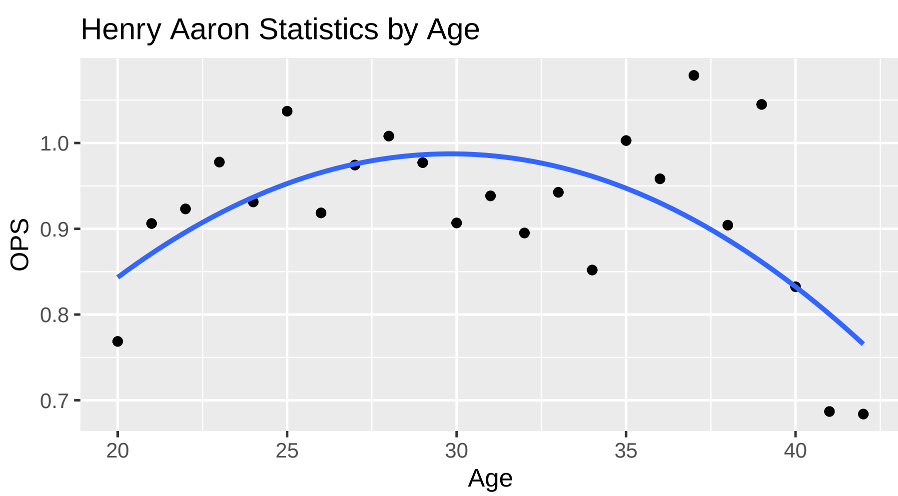
```r
# Get Aaron's stats.
aaron <- get_stats(aaronID)

aaron |>
  kable(digits = 3, caption = "Henry Aaron's statistics by age.")
```

Table 1: Henry Aaron's statistics by age.

playerID	Age	SLG	OBP	OPS
aaronha01	20	0.447	0.322	0.769
aaronha01	21	0.540	0.366	0.906
aaronha01	22	0.558	0.365	0.923
aaronha01	23	0.600	0.378	0.978
aaronha01	24	0.546	0.386	0.931
aaronha01	25	0.636	0.401	1.037
aaronha01	26	0.566	0.352	0.919
aaronha01	27	0.594	0.381	0.974
aaronha01	28	0.618	0.390	1.008
aaronha01	29	0.586	0.391	0.977
aaronha01	30	0.514	0.393	0.907
aaronha01	31	0.560	0.379	0.938
aaronha01	32	0.539	0.356	0.895
aaronha01	33	0.573	0.369	0.943
aaronha01	34	0.498	0.354	0.852
aaronha01	35	0.607	0.396	1.003
aaronha01	36	0.574	0.385	0.958
aaronha01	37	0.669	0.410	1.079
aaronha01	38	0.514	0.390	0.904
aaronha01	39	0.643	0.402	1.045
aaronha01	40	0.491	0.341	0.832
aaronha01	41	0.355	0.332	0.687
aaronha01	42	0.369	0.315	0.684

```
aaron |>
  ggplot(aes(x = Age,
             y = OPS)) +
  geom_point() +
  geom_smooth(method = "lm",
              se = FALSE,
              formula = y ~ poly(x, 2 , raw = TRUE)) +
  labs(title = "Henry Aaron Statistics by Age")
```



Fitting the Quadratic Model

```
fit <- lm(OPS ~ I(Age - 30) + I((Age - 30)^2), data = aaron)

A <- coef(fit)[1]
B <- coef(fit)[2]
C <- coef(fit)[3]

tibble(age.max = 30 - B / C / 2,
       max = A - B^2 / C / 4) |>
  kable(digits = 3)
```

age.max	max
29.817	0.987

There are a lot of things that can go wrong with this. What do you think about the fit in this instance?